

chRNA evidence review

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Helping scientists choose which chimeric-RNA junction to test next.

THE PROBLEM

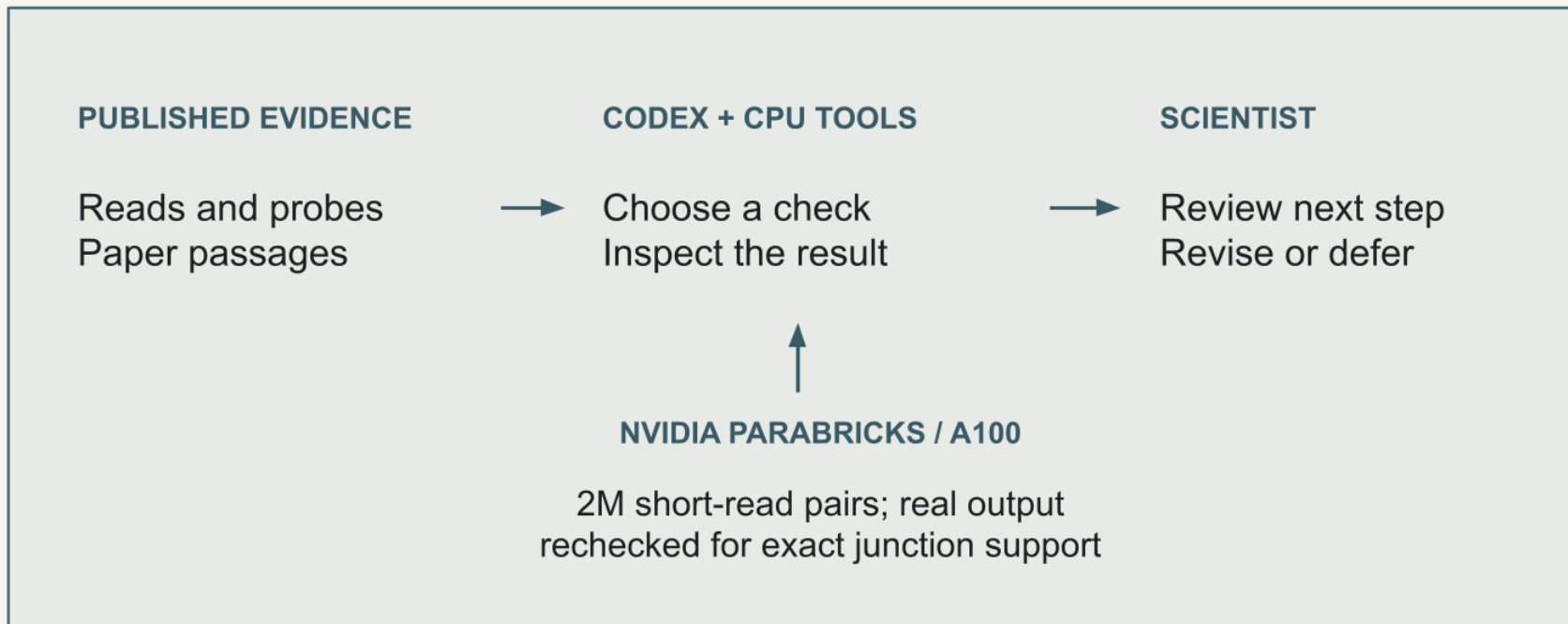
A gene-pair score does not identify a testable junction. Evidence is scattered across reads, probes and papers.

OUR APPROACH

Codex chooses evidence checks, reviews the results, and proposes a next step. A scientist can revise or defer it.

From candidate to scientist review

Codex executed 12 bounded tool calls across three cases on Brev. Every recorded output replays exactly.



A high score still needs a junction check

Psap-Lgals3

924 nt

Nearest read / probe endpoint discrepancy

Nearest read **chr14:47,622,058**

Designed probe **chr14:47,622,982**

Next: inspect source alignments before choosing the assay.

EVIDENCE

Three recorded cases; all 12 tool outputs replay exactly.

REAL-WORLD USE

A scientist checks the exact junction before choosing a validation assay.

KEY LIMITATION

Utility unmeasured.
Human review pending.

Direct source audit: Supplementary Tables 3/8 + GENCODE M28.
Descriptive discrepancy; long-read coordinate convention is unresolved.

Technical appendix

PROJECT LINKS

GitHub: [team-fusion-hackathon](#)

Offline: [demo/review/index.html](#)

Reviewed science: 76fb5de

Public repository; see release manifest

HOW TO REPRODUCE

Python 3.12 · uv · requirements.lock

python scripts/reproduce.py --download

python scripts/review/verify.py

Demo: python scripts/demo/serve.py

Core rebuild: identical predictions

TECHNICAL DETAILS

NVIDIA: Parabricks 4.7.1-1

A100 80 GB · 2M read pairs

85.49 s alignment; no speedup claim

OpenAI: Codex / gpt-6-astra

Recorded tool use; offline replay

Data: [Venezia et al., Nature \(2026\)](#)

GRCm39 / GENCODE M28

479 pairs; 5 gene-disjoint folds

Hi-C: AP 0.296 → 0.300 (n=401)

Δ AP 95% CI: -0.031 to +0.047

Python packages pinned; CPU replay

Cached Boltz2 2.2.1: low confidence